

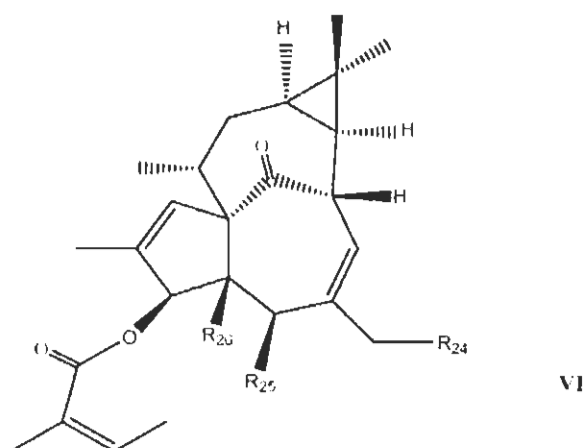
Amended Claims

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CLAIMS

1. A chemical agent obtainable from a plant of the Euphorbiaceae family ~~or a derivative or chemical analogue~~ or a pharmaceutically compatible salt thereof which
 5 chemical agent ~~derivative or chemical analogue~~ is a macrocyclic diterpene selected from compounds of the ingenane family represented by general formula (VI)



10 wherein:-

- R_{24} , R_{25} and R_{26} are independently selected from hydrogen, OH, R_{27} , R_{28} , F, Cl, Br, I, CN, OR_{27} , SR_{27} , $NR_{27}-R_{28}$, $N(-O)_2$, $NR_{27}-OR_{28}$, $ONR_{27}-R_{28}$, SOR_{27} , SO_2R_{27} , SO_2R_{27} , $SONR_{27}-R_{28}$, $SO_2NR_{27}-R_{28}$, $SO_2NR_{27}-R_{28}$, $P(R_{27})_3$, $P(-O)(R_{27})_3$, $Si(R_{27})_4$, $B(R_{27})_3$,
 15 $(C-X)R_{28}$ or $X(C-X)R_{28}$ where X is selected from sulfur, oxygen and nitrogen;

- R_{27} and R_{28} are each independently selected from C_1-C_{20} alkyl (branched or straight chained), C_1-C_{20} arylalkyl, C_1-C_{20} cycloalkyl, C_6-C_{14} aryl, C_1-C_{14} heteroaryl, C_1-C_{14} heterocycle, C_2-C_{10} alkenyl (branched or straight chained), C_2-C_{10} alkynyl (branched or
 20 straight chained), C_1-C_{10} heteroarylalkyl, C_1-C_{10} alkoxyalkyl, C_1-C_{10} haloalkyl, dihaloalkyl, trihaloalkyl, haloalkoxy, C_1-C_{10} alkyl which is substituted by CN, OR_{27} , SR_{27} ,

$\text{NR}_2\text{-R}_{27}$, $\text{N}(\text{O})$, $\text{NR}_2\text{-OR}_{27}$, $\text{ONR}_2\text{-R}_{27}$, SOR_{27} , SO_2R_{27} , SO_2R_{27} , $\text{SONR}_{27}\text{-R}_{27}$,
 $\text{SO}_2\text{NR}_{27}\text{-R}_{27}$, $\text{SO}_2\text{NR}_{27}\text{-R}_{27}$, $\text{P}(\text{R}_{27})_3$, $\text{P}(\text{O})(\text{R}_{27})_3$, $\text{Si}(\text{R}_{27})_4$, or $\text{B}(\text{R}_{27})_2$.

R_{27} and R_{27} are each independently selected from $\text{C}_1\text{-C}_{20}$ alkyl (branched or
 5 straight chained), $\text{C}_1\text{-C}_{20}$ arylalkyl, $\text{C}_3\text{-C}_{20}$ cycloalkyl, $\text{C}_6\text{-C}_{14}$ aryl, $\text{C}_1\text{-C}_{14}$ heteroaryl, $\text{C}_1\text{-C}_{14}$
 heterocycle, $\text{C}_2\text{-C}_{10}$ alkenyl (branched or straight chained), $\text{C}_2\text{-C}_{10}$ alkynyl (branched or
 straight chained), $\text{C}_1\text{-C}_{10}$ heteroarylalkyl, $\text{C}_1\text{-C}_{10}$ alkoxyalkyl, $\text{C}_1\text{-C}_{10}$ haloalkyl,
 dihaloalkyl, trihaloalkyl, haloalkoxy;

10 R_{27} is selected from R_{27} , R_{28} , CN , COR_{27} , CO_2R_{27} , OR_{27} , SR_{27} , $\text{NR}_{27}\text{-R}_{28}$,
 $\text{N}(\text{O})$, $\text{NR}_2\text{-OR}_{28}$, $\text{ONR}_2\text{-R}_{28}$, SOR_{27} , SO_2R_{27} , SO_2R_{27} , $\text{SONR}_{27}\text{-R}_{28}$, $\text{SO}_2\text{NR}_{27}\text{-R}_{28}$,
 $\text{SO}_2\text{NR}_{27}\text{-R}_{28}$, $\text{P}(\text{R}_{27})_3$, $\text{P}(\text{O})(\text{R}_{27})_3$, $\text{Si}(\text{R}_{27})_4$, $\text{B}(\text{R}_{27})_2$.

and which chemical agent, derivative or chemical analogue or pharmaceutically
 15 compatible salt thereof is capable of modulating PKC activity, PKC-dependent gene
 expression or PKC enzyme turnover for use in the treatment or prophylaxis of infection by
 a pathogenic organism or a potentially pathogenic organism in inflammatory condition in a
 subject and wherein said chemical agent, derivative or chemical analogue or
 pharmaceutically compatible salt thereof is for administration for a time and under
 20 conditions sufficient to ameliorate one or more symptoms associated with infection by said
 pathogenic organism in inflammatory condition or potentially pathogenic organism.

2. The chemical agent, derivative, chemical analogue or pharmaceutically compatible
 salt thereof for use as claimed in claim 1 wherein said chemical agent, derivative or
 25 chemical analogue or pharmaceutical compatible salt thereof exhibits a potency of agent
 (P_A) of >10 , wherein the $P_A = \sum I_i$ where I_i is a numerical value associated with a
 particular feature as listed below:-

Feature	Value
An ability to modulate PKC activity or effect	+1
An ability to induce bipolar dendritic activity	+1

Feature	Value
An ability to displace phorbol dibutyrate from binding to PKC	+1
An ability to induce respiratory burst in leucocytes	+1
An ability to stimulate phagocytosis in peripheral blood mononuclear cells	+1
Derived from a member of the Euphorbiaceae family	+1
Derived from <i>E. pepplus</i>	+3
Water extractable from the sap of <i>Euphorbia</i> sp.	+2
An ability to activate latent virus	+4
A lower tumor promotion activity than TPA/PMA	+2

or pharmaceutically acceptable salts of these.

3. The chemical agent, derivative, chemical analogue or pharmaceutically compatible salt thereof for use according to Claim 1 wherein R_{24} is H, O-Acetyl or OH.
4. The chemical agent, derivative, chemical analogue or pharmaceutically compatible salt thereof for use according to Claim 1 wherein R_{24} and R_{26} are OH.
5. The chemical agent, derivative, chemical analogue or pharmaceutically compatible salt thereof for use according to any one of Claims 1 to 4 wherein the plant is of the genus selected from *Acalypha*, *Acidoton*, *Actinostemon*, *Adelia*, *Adenocline*, *Adenocrepis*, *Adenophaedra*, *Adisca*, *Agrostistachys*, *Alchornea*, *Alchorneopsis*, *Alcinacanthus*, *Alcoceria*, *Aleurites*, *Amanoa*, *Andrachne*, *Angostyles*, *Anisophyllum*, *Antidesma*, *Aphora*, *Aporosa*, *Aporosella*, *Argythamnia*, *Astrococcus*, *Astrogynne*, *Baccanrea*, *Baliospermum*, *Bernardia*, *Beyeriopsis*, *Bischofia*, *Blachia*, *Blumeodondron*, *Bonania*, *Bradleya*, *Breynia*, *Brevniopsis*, *Briedelia*, *Burauavia*, *Caperonia*, *Caryodendron*, *Celtianella*, *Cephalocroton*, *Chaenotheca*, *Chaetocarpus*, *Chamaesyce*, *Cheilosa*, *Chiropetalum*, *Choriophyllum*, *Cicca*, *Chaoxylon*, *Cleidon*, *Cleistanthus*, *Chytia*, *Cnesmone*, *Cnidocolus*, *Coccoceras*, *Codiaeum*, *Coelodiscus*, *Conami*, *Conceveiba*, *Conceveibastrum*, *Conceveibum*, *Corythea*, *Croizatia*, *Croton*, *Crotonopsis*, *Crozophora*, *Cubanthus*, *Cumuria*, *Dactylostemon*, *Dalechampia*, *Dendroconusia*, *Diaspersus*, *Didymocistus*, *Dimorphocalyx*, *Discocarpus*,

- Ditaxis, Dodrechtia, Drypetes, Dysopsis, Elateriospermum, Endadenium, Endospermum, Erismanthus, Erythrocarpus, Erythrochilus, Eumecanthus, Euphorbia, Euphorbiadendron, Excoecaria, Flueggea, Galea, Garcia, Gavarretia, Gelonium, Giara, Givonia, Glochidion, Glochidionopsis, Glycydendron, Gymnanthes, Gymnosparia,*
- 5 *Haematospermum, Hendecandra, Hevea, Hieronima, Hieronyma, Hippocrepandra, Homalanthus, Hymenocardia, Janipha, Jatropha, Julocroton, Lasiocroton, Leioecarpus, Leonardia, Lepidanthus, Leucocroton, Mabca, Macaranga, Mallotus, Maritima, Mappa, Maprounea, Melanthesa, Mercurialis, Mettenia, Micrandra, Microdesmis, Microelus, Microstachy, Maocroton, Monadenium, Mozinga, Neoscortechinia, Omalanthus,*
- 10 *Omphalea, Ophellantha, Orbicularia, Ostodes, Oxydectes, Palengu, Pantadenia, Paradrypepes, Pausandra, Pedilanthus, Pera, Peridium, Petalostigma, Phyllanthus, Picrodendron, Picardia, Pilinophytum, Pimeleodendron, Piranhea, Platygyna, Plukenetia, Podocalyx, Poinsettia, Poraresia, Prosartema, Pseudanthus, Pycnocomia, Quadrasia, Reverchonina, Richeria, Richeriella, Ricinella, Ricinocarpus, Rottlera, Sagotia, Samwithia,*
- 15 *Sapium, Savia, Sclerocroton, Sebastiana, Securinega, Seneceldera, Senecildropsis, Serophyton, Siphonia, Spathiostemon, Spixia, Stillingia, Strophoblachia, Synadenium, Tetracoccus, Tetraplandra, Tetrorchidium, Thyrsanthera, Tithymalus, Trageia, Trewia, Trigonostemon, Tyria and Xylophylla.*
- 20 6. The chemical agent, ~~derivative, chemical analogue or pharmaceutically compatible salt thereof for use~~ according to Claim 5 wherein the plant is of the genus *Euphorbia*.
7. The chemical agent, ~~derivative, chemical analogue or pharmaceutically compatible salt thereof for use~~ according to Claim 6 wherein the species of *Euphorbia* is selected from
- 25 *Euphorbia aaron-rossii, Euphorbia abbreviata, Euphorbia acuta, Euphorbia alatoaulis, Euphorbia albicaulis, Euphorbia algomarginata, Euphorbia aliciae, Euphorbia alta, Euphorbia anacampseros, Euphorbia andromedae, Euphorbia angusta, Euphorbia anthonyi, Euphorbia antiquensis, Euphorbia apocynifolia, Euphorbia arabica, Euphorbia ariensis, Euphorbia arizonica, Euphorbia arkansana, Euphorbia arteagae, Euphorbia arundelana, Euphorbia astroites, Euphorbia atrococca, Euphorbia baselicis, Euphorbia batubaniensis, Euphorbia bergeri, Euphorbia bermudiana, Euphorbia bicolor, Euphorbia*
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- biformis*, *Euphorbia bifurcata*, *Euphorbia bilobata*, *Euphorbia biramensis*, *Euphorbia biuncialis*, *Euphorbia blepharostipula*, *Euphorbia blodgettii*, *Euphorbia boerhaavioides*, *Euphorbia boliviana*, *Euphorbia bracei*, *Euphorbia brachiata*, *Euphorbia brachycera*, *Euphorbia brandegeei*, *Euphorbia brittonii*, *Euphorbia caesia*, *Euphorbia calcicola*,
5 *Euphorbia campestris*, *Euphorbia candelabrum*, *Euphorbia capitellata*, *Euphorbia carmenensis*, *Euphorbia carunculata*, *Euphorbia cayensis*, *Euphorbia celastroides*, *Euphorbia chalicophita*, *Euphorbia chamaerhodos*, *Euphorbia chamaesula*, *Euphorbia chiapensis*, *Euphorbia chiogenoides*, *Euphorbia cinerascens*, *Euphorbia clarionensis*, *Euphorbia colimae*, *Euphorbia colorata*, *Euphorbia commutata*, *Euphorbia consoquitlae*,
10 *Euphorbia convolvuloides*, *Euphorbia corallifera*, *Euphorbia creberrima*, *Euphorbia crenulata*, *Euphorbia cubensis*, *Euphorbia cuspidata*, *Euphorbia cymbiformis*, *Euphorbia darlingtonii*, *Euphorbia defoliata*, *Euphorbia degeneri*, *Euphorbia deltoidea*, *Euphorbia dentata*, *Euphorbia depressa*, *Euphorbia dictyosperma*, *Euphorbia dictyosperma*, *Euphorbia dioeca*, *Euphorbia discoidalis*, *Euphorbia dorsiventralis*, *Euphorbia drumondii*,
15 *Euphorbia duclouxii*, *Euphorbia dussii*, *Euphorbia eanophylla*, *Euphorbia eggersii*, *Euphorbia eglandulosa*, *Euphorbia elata*, *Euphorbia enalla*, *Euphorbia eriogonoides*, *Euphorbia eriophylla*, *Euphorbia esculaeformis*, *Euphorbia espritensis*, *Euphorbia esula*, *Euphorbia excisa*, *Euphorbia exclusa*, *Euphorbia exstipitata*, *Euphorbia exstipitata*, *Euphorbia fendleri*, *Euphorbia filicaulis*, *Euphorbia filiformis*, *Euphorbia florida*,
20 *Euphorbia fruticulosa*, *Euphorbia garber*, *Euphorbia gaumerii*, *Euphorbia gerardiana*, *Euphorbia geyeri*, *Euphorbia glyptosperma*, *Euphorbia gorgonis*, *Euphorbia gracilior*, *Euphorbia gracillima*, *Euphorbia gradyi*, *Euphorbia graminea*, *Euphorbia graminea*, *Euphorbia grisea*, *Euphorbia guadalupana*, *Euphorbia guanarensis*, *Euphorbia gymnadenia*, *Euphorbia haematantha*, *Euphorbia hedyotoides*, *Euphorbia heldrichii*,
25 *Euphorbia helenae*, *Euphorbia helleri*, *Euphorbia helwigii*, *Euphorbia henricksonii*, *Euphorbia heterophylla*, *Euphorbia hexagona*, *Euphorbia hexagonoides*, *Euphorbia hinkleorum*, *Euphorbia hintonii*, *Euphorbia hirtula*, *Euphorbia hirta*, *Euphorbia hooveri*, *Euphorbia humistrata*, *Euphorbia hypericifolia*, *Euphorbia inundata*, *Euphorbia involuta*, *Euphorbia jaliscensis*, *Euphorbia jejuna*, *Euphorbia johnston*, *Euphorbia juttiae*,
30 *Euphorbia knuthii*, *Euphorbia lasiocarpa*, *Euphorbia lata*, *Euphorbia latazi*, *Euphorbia latericolor*, *Euphorbia laxiflora*, *Euphorbia lechenoides*, *Euphorbia ledienii*, *Euphorbia*

- leucophylla*, *Euphorbia lineata*, *Euphorbia linguiformis*, *Euphorbia longecornuta*,
Euphorbia longepetiolata, *Euphorbia longeramosa*, *Euphorbia longinsulicola*, *Euphorbia*
longipila, *Euphorbia lupulina*, *Euphorbia lurida*, *Euphorbia lycioides*, *Euphorbia*
macropodoides, *macvaughiana*, *Euphorbia manca*, *Euphorbia mandontana*, *Euphorbia*
5 *mangleti*, *Euphorbia mango*, *Euphorbia marylandica*, *Euphorbia mayana*, *Euphorbia*
melanadenia, *Euphorbia melanocarpa*, *Euphorbia meridensis*, *Euphorbia mertonii*,
Euphorbia mexiae, *Euphorbia microcephala*, *Euphorbia microclada*, *Euphorbia*
micromera, *Euphorbia misella*, *Euphorbia missurica*, *Euphorbia montana*, *Euphorbia*
montereyana, *Euphorbia multicaulis*, *Euphorbia multiformis*, *Euphorbia multinodis*,
10 *Euphorbia multiseta*, *Euphorbia muscicola*, *Euphorbia neomexicana*, *Euphorbia*
nephradenia, *Euphorbia niqeroana*, *Euphorbia oaxacana*, *Euphorbia occidentalis*,
Euphorbia odontodenia, *Euphorbia olivacea*, *Euphorbia olowatuana*, *Euphorbia*
ophthalmica, *Euphorbia ovata*, *Euphorbia pachypoda*, *Euphorbia pachyrhiza*, *Euphorbia*
padifolia, *Euphorbia palmeri*, *Euphorbia paludicola*, *Euphorbia parviflora*, *Euphorbia*
15 *pariskii*, *Euphorbia parryi*, *Euphorbia paxiana*, *Euphorbia pediculifera*, *Euphorbia*
peplidion, *Euphorbia peploides*, *Euphorbia pephus*, *Euphorbia pergamena*, *Euphorbia*
perlegna, *Euphorbia petaloidea*, *Euphorbia petaloidea*, *Euphorbia petrina*, *Euphorbia*
picachensis, *Euphorbia pilosula*, *Euphorbia pilulifera*, *Euphorbia pinariona*, *Euphorbia*
pinctorum, *Euphorbia pionosperma*, *Euphorbia platysperma*, *Euphorbia plicata*,
20 *Euphorbia poeppigii*, *Euphorbia poliosperma*, *Euphorbia polycarpa*, *Euphorbia*
polycnemoides, *Euphorbia polyphylla*, *Euphorbia portoricensis*, *Euphorbia portulacoides*
Euphorbia portulana, *Euphorbia preslii*, *Euphorbia prostrata*, *Euphorbia pteroneura*,
Euphorbia pycnanthema, *Euphorbia ramosa*, *Euphorbia rapulum*, *Euphorbia remyi*,
Euphorbia retroscabra, *Euphorbia revoluta*, *Euphorbia rivularis*, *Euphorbia robusta*,
25 *Euphorbia romosa*, *Euphorbia rubida*, *Euphorbia rubrosperma*, *Euphorbia rupicola*,
Euphorbia sanmartensis, *Euphorbia saxatilis* M. Bieb, *Euphorbia schizoloba*, *Euphorbia*
sclerocyathium, *Euphorbia scopulorum*, *Euphorbia senilis*, *Euphorbia serpyllifolia*,
Euphorbia serrula, *Euphorbia setiloba* Engelm, *Euphorbia sonorae*, *Euphorbia soohyi*,
Euphorbia sparsiflora, *Euphorbia sphacrosperma*, *Euphorbia syphilitica*, *Euphorbia*
30 *spruceana*, *Euphorbia subcoerulea*, *Euphorbia stellata*, *Euphorbia submamillaris*,
Euphorbia subpeltata, *Euphorbia subpubens*, *Euphorbia subreniforme*, *Euphorbia*

subrifoliata, *Euphorbia succedanea*, *Euphorbia tamaripasana*, *Euphorbia telephioides*,
Euphorbia tenuissima, *Euphorbia tetrapara*, *Euphorbia tirucalli*, *Euphorbia tomentella*,
Euphorbia tomentosa, *Euphorbia torralbasii*, *Euphorbia tovariensis*, *Euphorbia*
trachysperma, *Euphorbia tricolor*, *Euphorbia trojana*, *Euphorbia tuereckheimii*, *Euphorbia*
urceolaris, *Euphorbia umbellulata*, *Euphorbia undulata*, *Euphorbia vermiformis*,
Euphorbia versicolor, *Euphorbia villosa*, *Euphorbia violacea*, *Euphorbia whitei*,
Euphorbia xanti Engelm., *Euphorbia xytopoda* Greenm., *Euphorbia yayalesia* Urb.,
Euphorbia yungasensis, *Euphorbia zarayskiana* and *Euphorbia zinniflora*.

8. The chemical agent, derivative, chemical analogue or pharmaceutically compatible salt thereof for use according to Claim 7 wherein the species of *Euphorbia* is *Euphorbia pepus*.

9. The chemical agent, derivative, chemical analogue or pharmaceutically compatible salt thereof for use according to any one of claims 1 to 8 wherein the pathogenic organism or potentially pathogenic organism inflammatory condition is selected from: a bacteria, a virus, a fungus or an insect, tissue and/or organ transplant rejection, sepsis, acute respiratory distress syndrome (ARDS), asthma, trauma, oxidative stress, cell death, irradiation damage, ischemia, reperfusion, cancer, viral infection, autoimmune disease, rheumatoid arthritis, psoriasis, inflammatory bowel disease, glomerulonephritis, lupus, hepatitis, chronic hepatitis, juvenile diabetes, chronic non-suppurative thyroiditis, tuberculosis, syphilis, actinomycosis, sarcoidosis, amyloidosis, granulomatous thyroiditis, lymphocytic thyroiditis, Hashimoto's thyroiditis, invasive fibrous thyroiditis, Grave's disease, regional enteritis, Crohn's disease, granulomatous ileitis, ulcerative colitis, chorioretinal inflammatory syndrome, pancreatitis, synovitis of the hip,odynophagia, dysphagia, viral and bacterial pharyngitis, infectious mononucleosis, acute tonsillitis, peritonsillar abscess, ulcerative tonsillitis, lingual tonsillitis, Candidiasis, Epiglottitis, tracheobronchial inflammation, Ludwig's angina, idiopathic pulmonary fibrosis, interstitial lung disease, lichen planus, lichen sclerosus, abscess, meningitis, encephalitis, vasculitis, progressive multi-focal leukoencephalopathy, urticaria, spongiotic dermatitis, allergic contact dermatitis, dermatitis, chronic contact dermatitis, lichen simplex chronicus, atopic

dermatitis, erythema multiforme, stevens-johnson syndrome, toxic epidermal necrolysis, discoid lupus erythematosus and acne vulgaris.

5 10. The chemical agent or pharmaceutically compatible salt thereof for use according to any one of claims 1 to 9 wherein the symptom is itching.

10 141. ~~The chemical agent-derivative, chemical analogue or pharmaceutically compatible salt thereof~~ for use according to any one of Claims 1 to 109 wherein said chemical agent-
~~derivative or chemical analogue~~ is ingenol-3-angelate, 20-deoxy-inganol-3-angelate or 20-
O-acetyl-inganol-3-angelate or a pharmaceutically acceptable salt of these.

15 142. ~~The chemical agent, derivative, chemical analogue or pharmaceutically compatible salt thereof~~ for use according to claim 141 wherein said chemical agent-
~~chemical analogue~~ is ingenol-3-angelate or a pharmaceutically acceptable salt thereof.

20 123. ~~The chemical agent- derivative, chemical analogue or pharmaceutically compatible salt thereof~~ for use according to any one of Claims 1 to 123 wherein said chemical agent-
~~derivative or chemical analogue or pharmaceutically compatible salt thereof~~ is provided in the form of a composition consisting of
~~comprising a pharmaceutically- or cosmetically-~~ acceptable carrier and said chemical agent or pharmaceutically compatible salt thereof.

25 124. ~~The chemical agent-derivative, chemical analogue or pharmaceutically compatible salt thereof~~ for use according to Claim 132 wherein said carrier is selected from β -alanine
betaine hydrochloride and t-4-hydroxy-N,N-dimethylproline.

145. Use of a chemical agent obtainable from a plant of the Euphorbiaceae family or a
derivative or chemical analogue or a pharmaceutically acceptable salt thereof wherein said
chemical agent, derivative, chemical analogue or pharmaceutically compatible salt thereof
is as defined in any one of claims 1 to 8 or 101 to 134 for the manufacture of a medicament
5 for the treatment or prophylaxis of infection by a pathogenic organism or a potential
pathogenic organism ~~an inflammatory condition~~ in a subject and wherein said chemical
agent, derivative or chemical analogue or pharmaceutically compatible salt thereof is for
administration for a time and under conditions sufficient to ameliorate one or more
symptoms associated with infection by said pathogenic organism or potential pathogenic
10 organism ~~inflammatory condition~~.

156. Use as claimed in claim 145 wherein said condition is as defined in claim 9 or 10.